www.tntextbooks.in



Figure 3.21 Tangent Galvanometer

Tangent law

When a magnetic needle or magnet is freely suspended in two mutually perpendicular uniform magnetic fields, it will come to rest in the direction of the resultant of the two fields.

Let B be the magnetic field produced by passing current through the coil of the tangent Galvanometer and B_H be the horizontal component of earth's magnetic field. Under the action of two magnetic fields, the needle comes to rest making angle θ with B_H, such that

$$B = B_H \tan \theta \qquad (3.29)$$

Construction

Tangent Galvanometer (TG) consists of copper coil wounded on a non-magnetic circular frame. The frame is made up of brass or wood which is mounted vertically on a horizontal base table (turn table) with three levelling screws as shown in Figure 3.22. The TG is provided with two or more coils of different number of turns. Most of the equipment we use in laboratory consists of 2 turns, 5 turns and 50 turns which are of different thickness and are used for measuring currents of different strengths.

At the center of turn table, a small upright projection is seen on which compass box (also known as magnetometre box) is placed. Compass box consists of a small magnetic needle which is pivoted at the center, such that arrangement shows the center of both magnetic needle and circular coil exactly coincide. A thin aluminium pointer is attached to the magnetic needle normally and moves over circular scale. The circular scale is divided into four quadrants and graduated in degrees which are used to measure the deflection of aluminium pointer on a circular degree scale. In order to avoid parallax error in measurement, a mirror is placed below the aluminium pointer.

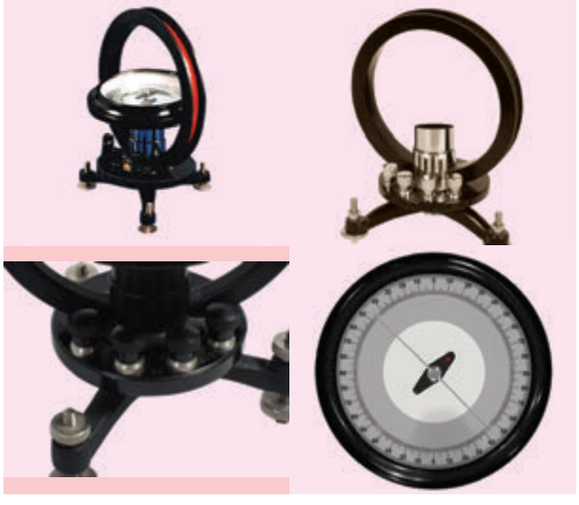


Figure 3.22 Tangent Galvanometer and its parts

Precautions

1. All the nearby magnets and magnetic materials are kept away from the instrument.
2. Using spirit level, the levelling screws at the base are adjustedso that the small

UNIT-3(XII-Physics_Vol-1).indd 147 04-03-2019 10:33:48